

basic_istream_view::iterator should not be copyable

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Proposed change

In [P1027] we discussed why iterators over a single-pass resource such as a stream should not be copyable. `basic_istream_view::iterator` proposed by [P1035] is one such iterator (and the only one proposed for C++20), we therefore propose to make it non-copyable.

Applicability

This paper depends on both [P1027] and [P1035] being accepted by LWG. They have been accepted by LEWG in Kona 2019.

Wording

The wording changes are to be applied on top of the wording changes proposed by [P1035]

◆ **Class template `basic_istream_view::iterator`** **[`range.istream.iterator`]**

```
namespace std::ranges {  
template<class Val, class CharT, class Traits>  
class basic_istream_view<Val, CharT, Traits>::iterator { // exposition only  
    public:  
        using iterator_category = input_iterator_tag;  
        using difference_type = ptrdiff_t;  
        using value_type = Val;  
  
        iterator() = default;  
        constexpr explicit iterator(basic_istream_view& parent) noexcept;  
  
        iterator(const iterator&) = delete;  
        iterator(iterator&&) = default;  
  
        iterator& operator=(const iterator&) = delete;  
};
```

```

    iterator& operator=(iterator&&) = default;

    iterator& operator++();
    void operator++(int);

    Val& operator*() const;

    friend bool operator==(iterator x, default_sentinel);
    friend bool operator==(default_sentinel y, iterator x);
    friend bool operator!=(iterator x, default_sentinel y);
    friend bool operator!=(default_sentinel y, iterator x);
    private:
    basic_istream_view<Val, CharT, Traits>* parent_ = nullptr; // exposition only
};

```

References

- [P1027] Corentin Jabot *Movability of single-pass iterators* <https://wg21.link/P1027>
- [P1035] Christopher Di Bella, Casey Carter, Corentin Jabot *Input range adaptors*
<https://wg21.link/P1035>